VDM

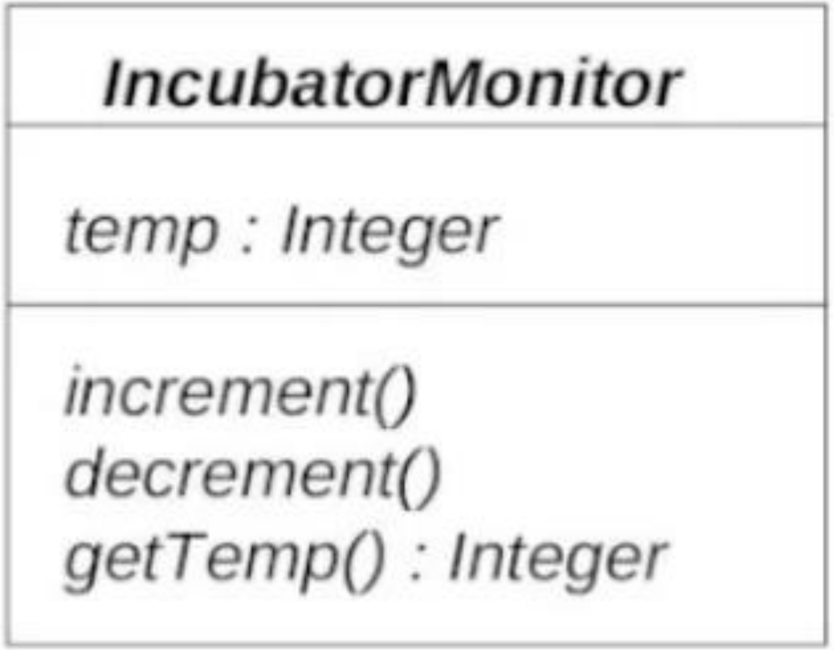
The Incubator case study

The temperature of the incubator needs to be carefully controlled and monitored;

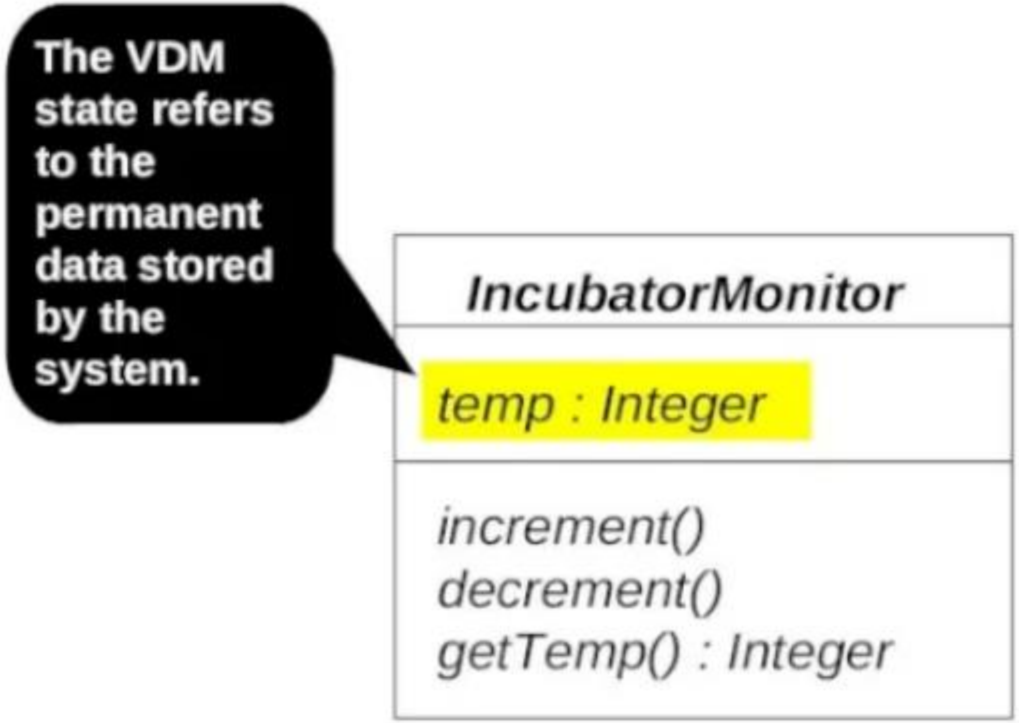
Safety requirements :

-10 Celsius ⟵ TEMPERATURE ⟶ +10 Celsius

The UML specification



Specifying the ‘state’ in VDM-SL



In VDM-SL we use mathematical types

TYPES AVAILABLE IN VDM

▯ : natural numbers (positive whole numbers)

▯$_1$: natural numbers excluding zero

▯: integers (positive and negative whole numbers)

▯: real numbers (positive and negative numbers that can include a fractional part)

▯ : boolean values (true or false)

Char : the set of alphanumeric characters

UML

IncubatorMonitor

temp : Integer

increment()
decrement()
getTemp() : Integer

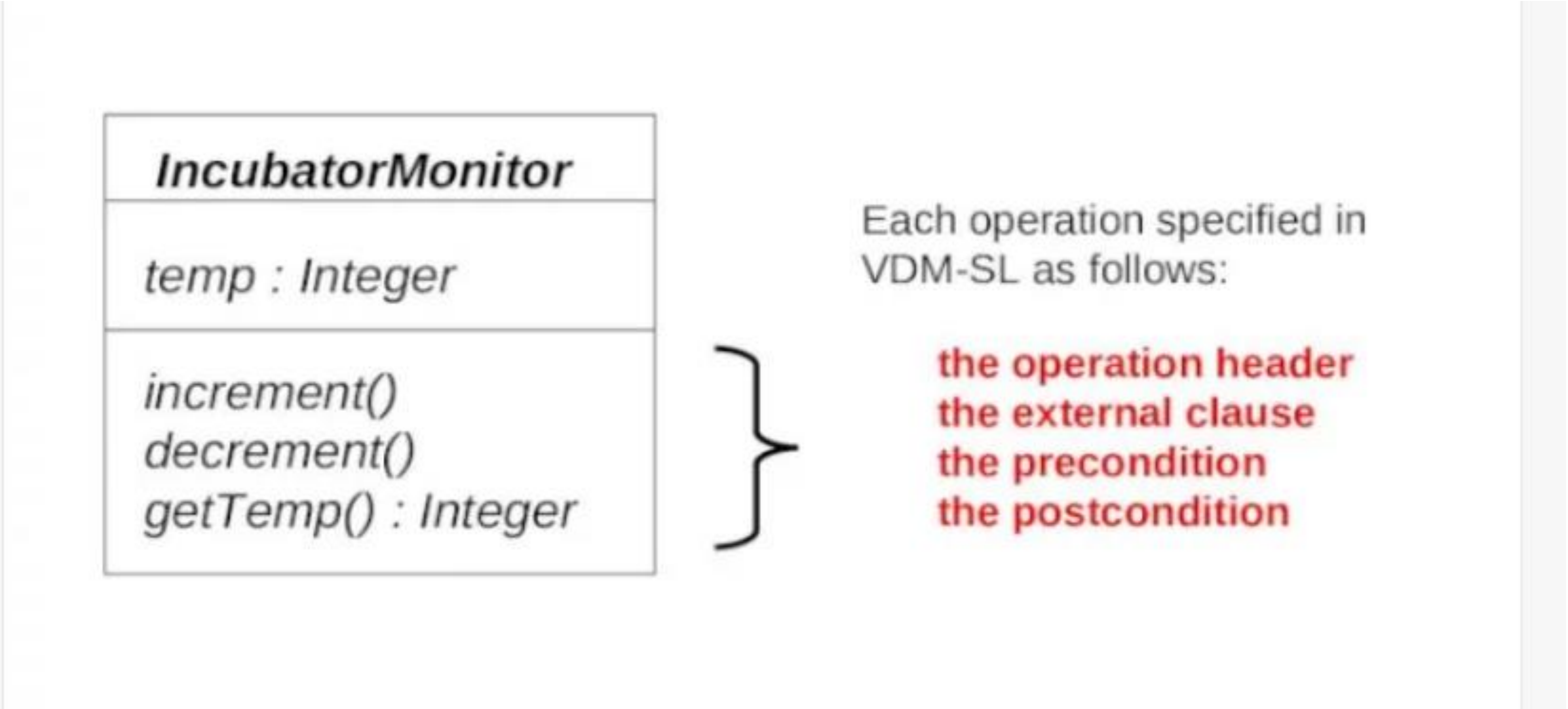
VDM-SL

state *IncubatorMonitor* **of**

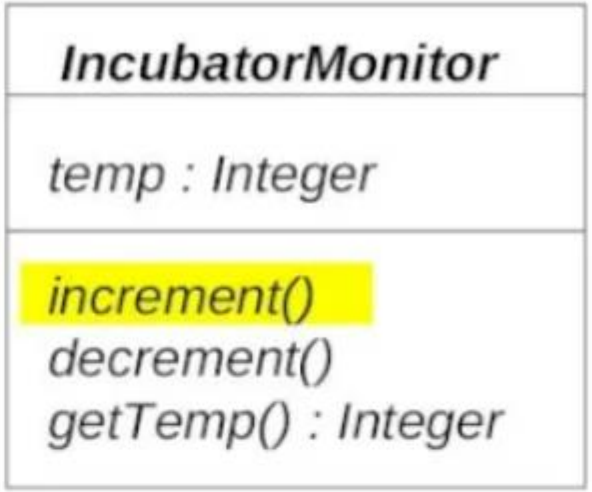
temp : ▯

end

Operations in VDM SL



Increment Operation



increment()

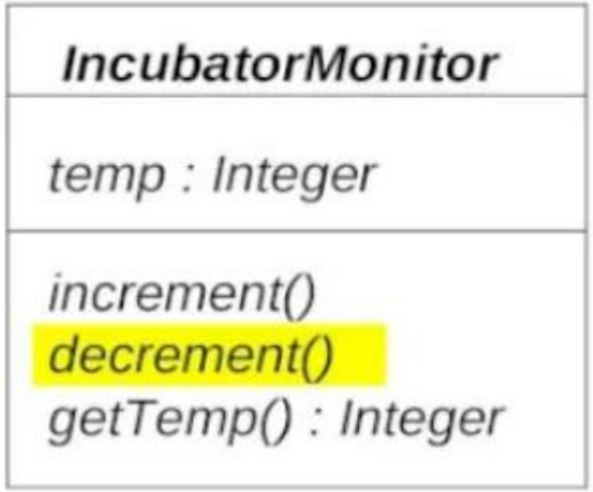
ext **wr** $temp$: ▯

pre $temp < 10$

post $temp > \overline{temp}$

$\overline{temp} + 1 = temp$

Decrement Operation



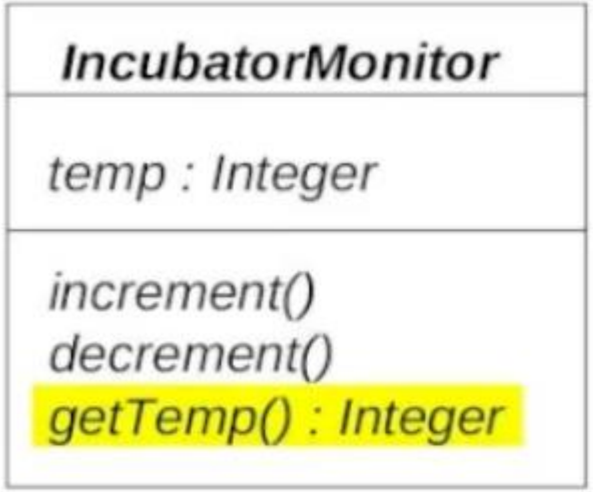
decrement()

ext **wr** $temp$: ▯

pre $temp > -10$

post $temp = \overline{temp} - 1$

Get Temperature



getTemp( ) *currentTemp* : $\mathbb{Z}$

ext **rd** *temp* : $\mathbb{Z}$

pre TRUE

post *currentTemp* = *temp*